PRODUCT REQUIREMENTS DOCUMENT · ARCHITECTURE

EmbeddedLend: API-First SME Underwriting Layer

The system contract for the unbuilt bridge — consent-based mobile-money underwriting on top of BNR Open Finance (data rail) and eKash (money rail). API-only scope; UI deferred.

EmbeddedLend (rename at PRD stage)

Rwanda → East African Community

PRD (architecture) → MVP slice flagged → Demo build

v1.0 · August 2026

Problem Statement, Qianhe Autopsy

Abstract

EXECUTIVE SUMMARY

EmbeddedLend is an API-first SME underwriting service for the "missing middle" of Rwanda's credit market. It converts consented mobile-money cash-flow into defensible, explainable loan decisions, sitting on top of the two national rails: BNR Open Finance (the data rail, voluntary phase now) and eKash (the money rail, live July 2026).

This PRD specifies the full architecture — a layered stack from consent and data access, through deterministic signal extraction and scoring, to loan operations and eKash money movement, with compliance and audit as a vertical slice. The MVP slice (flagged throughout) is the "one livable room" that proves the core bet: one consent flow, one data source, the scoring engine, the eKash connector, and the core endpoints. UI productization is explicitly out of scope; a demo interface is built separately as an outreach artifact.

Contents

1. Goals & Non-Goals

Goals

- Deliver an **API-first SME underwriting service** enabling licensed lenders to originate, underwrite, and service business loans for the "missing middle" — MSMEs too informal for collateral-based bank lending and too business-like for consumer micro-loans.
- Turn consented **mobile-money cash-flow** into a **defensible, explainable loan decision** — the unbuilt bridge between the data rail (BNR Open Finance) and the money rail (eKash).
- Make the regulator comfortable with **more lending, not less**: consent-first data pulls, audit trails, explainable decisions, data residency.

Non-Goals (explicitly out of scope)

- **No UI/UX productization** in this PRD — no lender portal screens, no onboarding flows. UI is deferred pending primary research on the player landscape; a demo interface is built separately as an outreach artifact.
- No direct consumer lending, no deposits, no holding a loan book (the Qianhe lesson — we never take credit risk; the lender remains the regulated entity).
- No ML model training or black-box scoring in the MVP — deterministic, transparent scoring first.
- No multi-bureau, multi-rail integration in the MVP.

2. Personas (Primary — who pays)

Persona	Description	Consumes
NDFSP (non-deposit-taking financial service provider)	One of 80+ licensed digital lenders in Rwanda (e.g. Zamuka, IWACU, EXUUS). Tech-poor, small, needs plug-and-play origination + scoring.	Full stack: consent, data pull, score, decision, loan, disbursement, repayment
MFI / SACCO	Extends credit to women and micro-enterprises; runs on field-officer apps and paper.	Scoring API + decision API (brings own distribution)
Vertical SaaS platform	E-commerce, logistics, agritech, retail POS serving merchants (e.g. Kayko corridor). Wants embedded working capital without becoming a lender.	Embedded credit flow: score + decision + loan + eKash disbursement inside their product

Secondary (future): regional banks preferring white-label APIs; digital lenders expanding into Rwanda.

3. Architecture Overview

A layered, API-first microservice integration layer. The lender's core banking system stays the system of record; our services sit beside it. At a glance, the whole system: consented data flows in on the left, is scored and decided in the underwriting core, a policy gate approves, a loan is disbursed over eKash, and repayment history loops back to improve future scores.

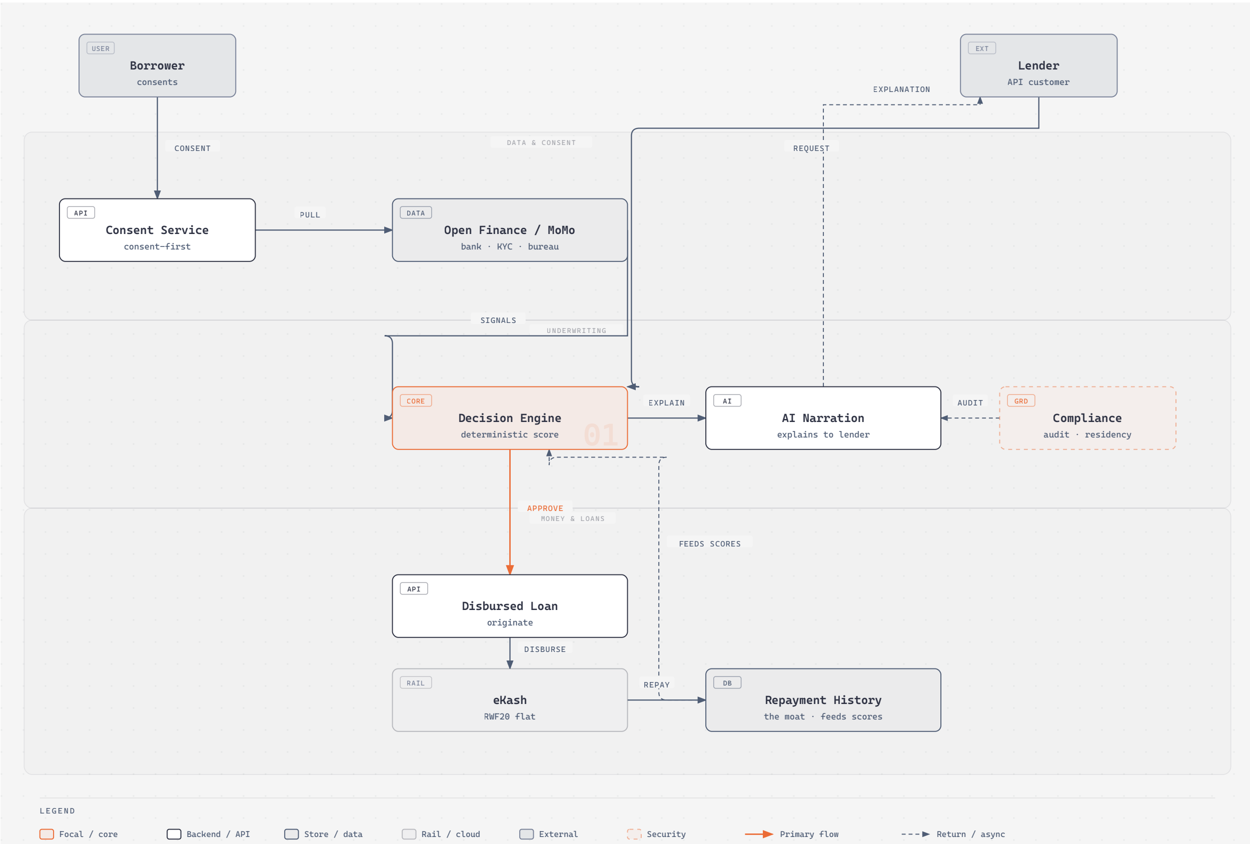

Figure 1 — EmbeddedLend, the system at a glance (editorial overview)

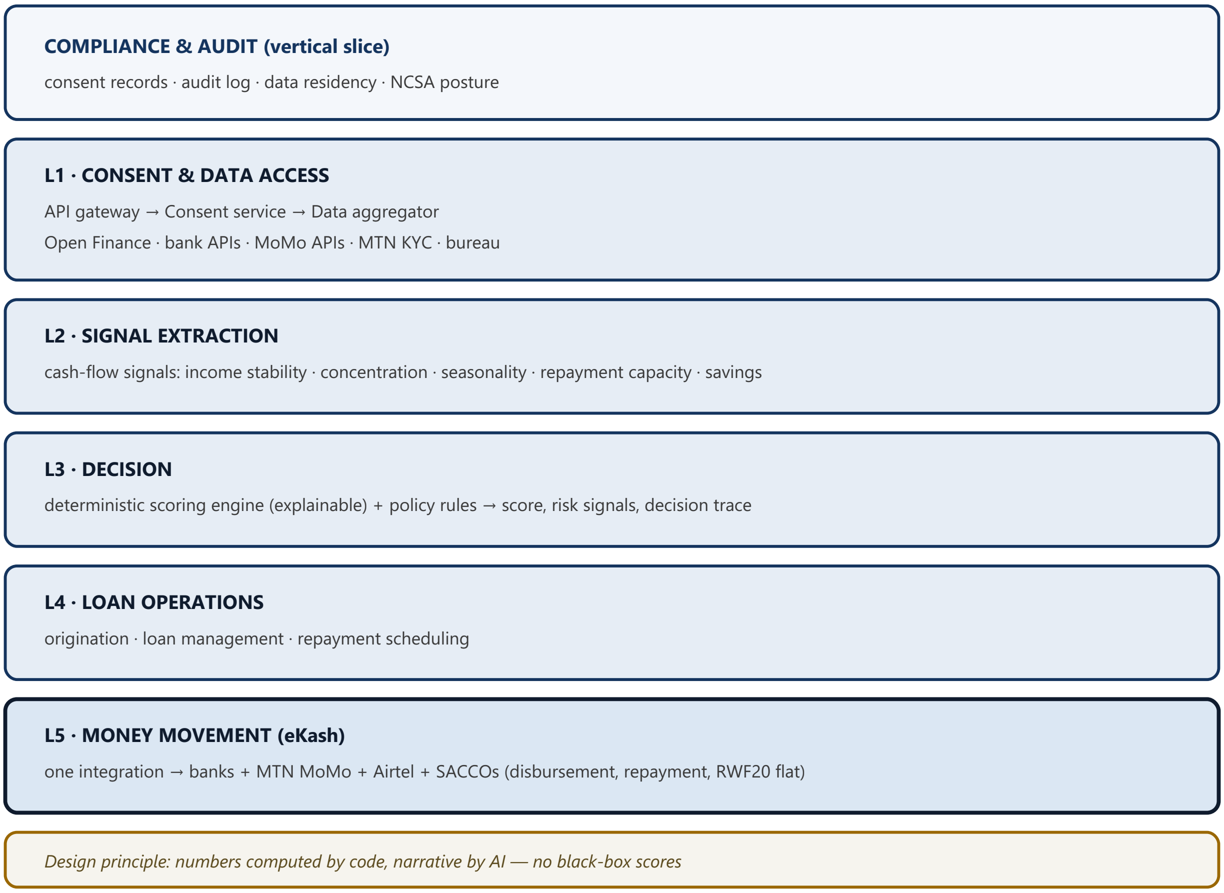

Figure 2 — EmbeddedLend layered architecture (MVP slice shaded in Figure 4)

4. Data Flow (Borrower Journey)

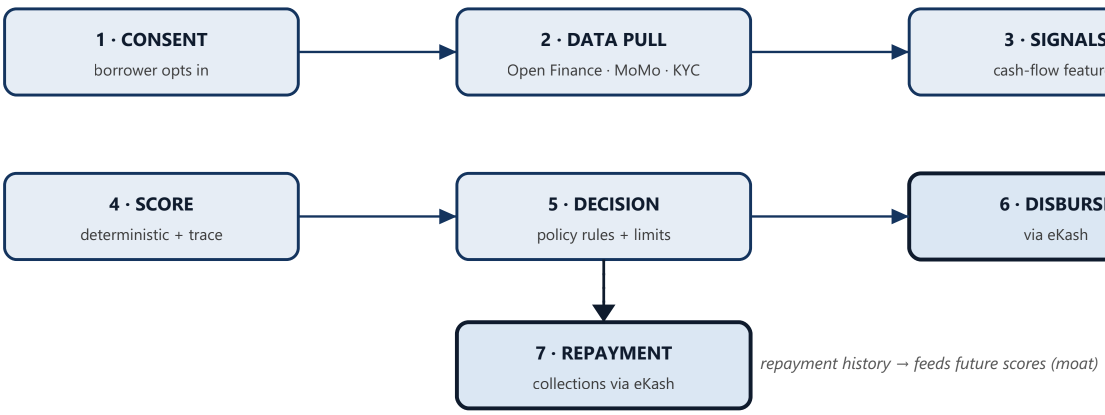

Figure 3 — Data flow: consent → data → signals → score → decision → eKash disbursement → repayment

5. Component Map

Component	Responsibility

API Gateway	AuthN/AuthZ, rate limiting, tenant isolation per lender
Consent Service	Capture, record, revoke consent; audit-gated
Data Aggregator	Normalize Open Finance / bank / MoMo / KYC / bureau data into one schema
Signal Extractor	Compute cash-flow signals (deterministic)
Scoring Engine	Deterministic, explainable score + risk signals
Decision Service	Policy rules, limits, decision trace
Loan Manager	Origination, loan lifecycle, repayment scheduling
eKash Connector	Disbursement + repayment over the national rail
Audit Store	Tamper-proof decision + consent audit trail

6. API Surface Sketch

Endpoint	Purpose
`POST /v1/consent`	Create borrower data-sharing consent
`GET /v1/data/summary`	Fetch consented, aggregated data + cash-flow signals
`POST /v1/score`	Compute deterministic score + risk signals
`POST /v1/decisions`	Apply policy rules → decision + trace
`POST /v1/loans`	Originate a loan
`POST /v1/disbursements`	Disburse via eKash
`POST /v1/repayments`	Record/collect repayment via eKash

7. Compliance Architecture

- **Consent-first** data pulls with purpose limitation (Law 058/2021 + GDPR-style rights: explainability, correction, portability).
- **NCSA posture** — data processor/controller certification path per the Open Finance roadmap.
- **Data residency** in jurisdiction; tamper-proof decision traces; full audit log.
- **Explainability** — every decision carries its risk signals and reasoning; the "numbers computed, narrative by AI" principle keeps the LLM as narrator, never the decider.

8. MVP Slice (Flagged — what we build first)

MVP — THE "ONE LIVABLE ROOM"

The minimum cut that proves the core bet: a lender can plug in, get a borrower's mobile-money data, get a defensible score, and disburse/repay via eKash.

In MVP	Out of MVP (later)
Consent flow (one)	Loan-management dashboards
One data source (mobile-money cash-flow)	Multiple bureaus
Scoring engine (deterministic, explainable)	KYC vendor integrations
Decision service + trace	ML model training
eKash connector (disburse + repay)	Fancy reporting / analytics
Core endpoints: consent, data/summary, score, decisions, loans, disbursements, repayments	Multi-country expansion

MVP success metric: a pilot lender originates a real business loan to a thin-file borrower using consented mobile-money data, with a defensible decision trace — proving willingness-to-pay and data-layer feasibility.

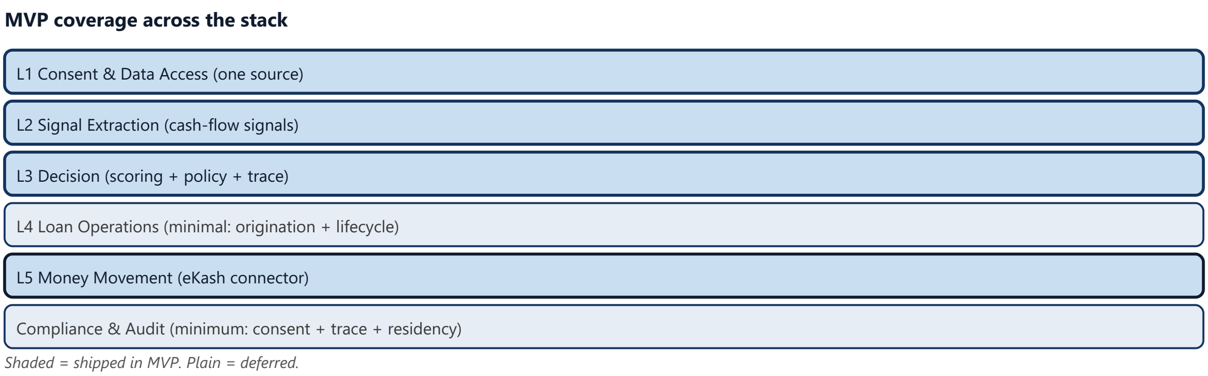

Figure 4 — MVP slice flagged across the architecture

9. Tech Stack

- **Backend:** Node/TypeScript, Postgres
- **Infra:** API-first microservices, containerized
- **Money rail:** eKash (one integration → whole ecosystem)
- **Data rail:** Open Finance / bank / MoMo consent APIs; MTN KYC API

- **Decision:** deterministic scoring engine (pure compute) + optional LLM narration layer (config at deploy time)

10. Evidence Ledger (PRD-specific)

Claim	Source	Grade
eKash = one integration for the whole ecosystem (banks, MTN MoMo, Airtel, SACCOs, MFIs), RWF20 flat, RWF10m cap	New Times, African Business, RISA (Jul 2026)	✅ **Observed**
eKash is a payments rail only — no credit scoring/loan decisioning	Launch coverage	✅ **Observed**
BNR Open Finance objective #1 = alternative scoring for MSMEs; voluntary Phase 1 now, mandatory Phase 2 (2026-28), central platform Phase 4 w/ private partner	BNR Open Finance Position Paper	✅ **Observed**
MTN KYC API exists, consent-gated, use case "digital lending"	MTN Rwanda	✅ **Observed**
BK Open API (Mar 2026) enables payments/credit access/data exchange	KT Press	✅ **Observed**
80+ licensed NDFSPs; Jali Finance: "credit reference systems remain a structural challenge"	BNR list; allAfrica (Apr 2026)	✅ **Observed**
Deterministic-compute + LLM-narration pattern validated in finance (FinRobot)	FinRobot repo / AI4Finance	🟡 **Secondary**
Willingness-to-pay by Rwandan lenders	No interviews yet	❓ **Assumption**

Method: architecture derived from the problem statement (problem-embeddedlend.md); facts graded in Section 10. The MVP slice is the first implementable increment; UI productization and primary research on the player landscape run in parallel. Open question: lender willingness-to-pay, to be validated in the Field stage.